Surveillance Pipeline Architecture & Technical Verification

1. End-to-End Processing Architecture

Stage 1: Spatial Gating & Motion Analysis: Evaluates native frames within the calibrated mechanical boundary [295, 20, 1490, 780] to filter irrelevant terrace activity.
Stage 2: Lossless Bitstream Extraction: Executes direct FFmpeg HEVC packet copying with enforced ±60-second pre- and post-event contextual buffers.
Stage 3: Deep Feature Extraction: Crops faces passing confidence >= 0.80 and quality score Q >= 40.0, extracting L2-normalized 512-D ArcFace vectors.
Stage 4: Unsupervised Spatial-Temporal Identity Clustering: Groups vectors via DBSCAN (cosine distance threshold 0.38) to build objective master personas.

2. Calibrated Native 3K Spatial Geometry & Equipment Zones

To ensure exact bounding box alignment between raw video files and NVR monitoring stations, coordinates are mapped directly onto the native 3K (2880 x 1620) plane:

Equipment Hazard Zone	Native 3K Coordinates [X1, Y1, X2, Y2]	Dimensions	Criticality Level
Stage 1 Observation Perimeter	**[295, 20, 1490, 780]**	1195 x 760 px	Boundary Gating Mask
Zone 1A (Skid Multiport Valve)	**[1080, 370, 1210, 500]**	130 x 130 px	**CRITICAL HAZARD**
Zone 1B (Filter Head Multiport Valves)	**[810, 440, 990, 580]**	180 x 140 px	**CRITICAL HAZARD**
Zone 2 (Ultimate Power Switch)	**[995, 340, 1095, 495]**	100 x 155 px	**SEVERE HAZARD**
Zone 3 (RO Skid Controls & Walkway)	**[1080, 320, 1420, 790]**	340 x 470 px	ELEVATED MONITORING

3. Temporal Integrity & Boundary Bridging Engineering

1. Continuous Rolling Temporal Windows: Eliminates activity window timeouts by maintaining a sliding state across long 45-minute CCTV segments, preventing missed re-entry events.
2. Multi-Segment Boundary Stitching: Enforces ±60-second contextual safety buffers that automatically bridge adjacent video files, capturing full pre-incident approach and post-incident egress.
3. ISO 8601 Wall-Clock Normalization: All detection queries enforce ORDER BY timestamp ASC using absolute wall-clock timestamps, eliminating intra-clip relative index interleaving.

4. Lossless Bitstream Extraction Performance Benchmark

• Direct FFmpeg bitstream extraction (-c copy) bypasses costly CPU/GPU re-encoding and preserves 100% bit-for-bit forensic image quality.
• **Benchmark Throughput:** Successfully extracted **179 pristine MP4 incident clips (8.20 GB) in 108.7 seconds** directly from read-only storage (**>250x realtime speed**).